

REVIEW - 1 DOCUMENT

Hybrid RAG Framework for Enterprise Knowledge Management

1. Project Title: Hybrid RAG Framework for Enterprise Knowledge Management

2. Problem Statement

Enterprise organizations maintain large collections of annual reports, financial reports, technical documents, product documentation and other corporate knowledge. Manually locating relevant information is time-consuming, while conventional keyword search may not capture the semantic meaning of a user's question. Retrieval-Augmented Generation (RAG) connects an LLM to an external knowledge base, but the quality of the generated answer depends on retrieving relevant evidence. This project proposes a practical Hybrid RAG framework that combines dense semantic retrieval and lexical keyword retrieval to improve the relevance and coverage of enterprise information provided to the generation model.

3. Related Literature → Limitations → Research Gaps → Objectives

This project follows the sequence required for research-gap identification: first, the related literature is reviewed; second, limitations or scope constraints are identified from the studies; third, common limitations are synthesized into specific research gaps; finally, each research objective is formulated to directly address one or more research gaps.

4. Literature Survey – 10 Related Research Papers

Note: The limitation statements below distinguish between (a) limitations or open issues explicitly discussed by the authors and (b) scope limitations inferred from what the study evaluates. Where a paper does not claim a method is universally limited, the wording uses the narrower and defensible description 'scope limitation'.

No.	Research paper	Model / framework	Outcome	Limitation / scope relevant to this research	Dataset(s) + link
1	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (Lewis et al., 2020)	RAG	Combines a seq2seq generator with a dense vector index; improves knowledge-intensive QA and factual generation.	The study uses a Wikipedia knowledge source and focuses on knowledge-intensive NLP benchmarks; enterprise-document retrieval and lexical+dense hybrid retrieval are outside its evaluated scope.	Natural Questions, WebQuestions, CuratedTREC, SQuAD; https://arxiv.org/abs/2005.11401
2	REALM: Retrieval-Augmented Language Model Pre-Training (Guu et al., 2020)	REALM	Introduces a latent knowledge retriever and demonstrates improved open-domain QA accuracy.	Knowledge is integrated through pre-training/fine-tuning with a large corpus such as Wikipedia; this is substantially more training-intensive than a practical enterprise RAG	Natural Questions, WebQuestions, CuratedTREC; Wikipedia; https://arxiv.org/abs/2002.08909

				pipeline and is not enterprise-specific.	
3	Dense Passage Retrieval for Open-Domain Question Answering (Karpukhin et al., 2020)	DPR	Dense dual-encoder retrieval outperformed a strong BM25 baseline by 9–19 percentage points in top-20 passage retrieval accuracy across evaluated benchmarks.	The approach demonstrates the effectiveness of dense retrieval, but the study is open-domain and does not evaluate a practical enterprise Hybrid RAG system combining dense and lexical retrieval.	Natural Questions, TriviaQA, WebQuestions, CuratedTREC, SQuAD; https://aclanthology.org/2020.emnlp-main.550/
4	Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering (Izacard & Grave, 2021)	Fusion-in-Decoder (FiD)	Encodes retrieved passages independently and fuses them in the decoder, achieving strong open-domain QA results.	The approach depends on retrieved passages and increases computation as more passages are processed; its evaluation is open-domain rather than enterprise-specific.	Natural Questions, TriviaQA, SQuAD; https://aclanthology.org/2021.eacl-main.74/
5	Precise Zero-Shot Dense Retrieval without Relevance Labels (Gao et al., 2023)	HyDE	Uses an LLM-generated hypothetical document to improve zero-shot dense retrieval and reports strong results across retrieval tasks and languages.	The hypothetical document is synthetic and can contain hallucinated information; the approach remains a dense-retrieval method rather than an enterprise lexical+dense hybrid pipeline.	BEIR and MIRACL; https://aclanthology.org/2023.acl-long.99/
6	Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection (Asai et al., 2023)	Self-RAG	Allows the model to retrieve on demand and critique retrieved evidence and generated outputs using reflection tokens; improves factuality on several tasks.	Requires specialized training and a more complex generation/retrieval mechanism; the reported work is not focused on a lightweight enterprise document-management deployment.	PubHealth, ARC-Challenge, TriviaQA, PopQA, ASQA, FactScore; https://arxiv.org/abs/2310.11511
7	Corrective Retrieval Augmented Generation	CRAG	Adds a retrieval evaluator and corrective actions; experiments show	The framework introduces an additional retrieval evaluator	Four datasets covering short- and long-form generation tasks; https://arxiv.org/abs/2401.15884

	(Yan et al., 2024)		improved performance when retrieval quality is poor.	and can use web search as an extension; this adds architectural complexity compared with a simple enterprise hybrid retriever.	
8	From Local to Global: A Graph RAG Approach to Query-Focused Summarization (Edge et al., 2024)	GraphRAG	Builds an entity graph and community summaries; improves comprehensiveness and diversity for global questions over large private text corpora.	Graph construction and pre-generated community summaries add preprocessing and computational complexity; the method targets global sensemaking and is more complex than the proposed lightweight hybrid retrieval pipeline.	Private text corpora / 1-million-token-range evaluation; https://arxiv.org/abs/2404.16130
9	ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems (Saad-Falcon et al., 2024)	ARES	Automates evaluation of context relevance, answer faithfulness and answer relevance using synthetic data, lightweight judges and a small set of human annotations.	ARES is an evaluation framework rather than an enterprise retrieval architecture; its experiments are based on knowledge-intensive benchmark tasks, so enterprise-specific evaluation still requires an appropriate domain dataset.	KILT, SuperGLUE and AIS tasks; https://aclanthology.org/2024.naacl-long.20/
10	EKRAG: Benchmark RAG for Enterprise Knowledge Question Answering (Yu et al., 2025)	EKRAG	Introduces an enterprise RAG benchmark covering corporate documents such as product releases, technical blogs and financial reports; evaluates retrieval strategies and answer correctness, relevance and faithfulness.	The work is primarily an enterprise benchmark and evaluation study. It provides evidence and guidance for enterprise RAG, but the exact practical system proposed here—dense FAISS retrieval combined with BM25 over the project's three corporate collections—is not	EKRAG enterprise QA benchmark; corporate product releases, technical blogs, financial reports; https://aclanthology.org/2025.knowledgenlp-1.13/

				the same experimental setup.	
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5. Cross-Paper Limitation Analysis

The limitations and scope boundaries above can be synthesized into the following recurring issues:

- Enterprise specificity: Several foundational RAG and dense-retrieval studies evaluate open-domain or benchmark knowledge rather than heterogeneous enterprise document collections.
- Retrieval coverage: Dense retrieval is strong for semantic similarity, while enterprise queries may also contain exact names, numbers, dates, product codes and technical terminology that benefit from lexical matching.
- Architectural complexity: Advanced approaches such as Self-RAG, CRAG and GraphRAG introduce additional training, evaluation, correction, or graph-building components that may increase implementation complexity.
- Controlled comparison: A clear enterprise experiment comparing dense-only RAG with dense-plus-lexical Hybrid RAG under the same documents, questions and generator is still a useful practical research direction.
- Evaluation: Enterprise RAG needs evaluation of both retrieval quality and generated-answer quality rather than relying on a single generic accuracy number.

6. Research Gap Identification

Research Gap	How it is derived from the literature	Why it matters to our project
Research Gap 1 – Enterprise-focused hybrid retrieval	Foundational RAG/DPR/FiD studies establish dense retrieval and generation but are mainly open-domain; EK-RAG establishes enterprise RAG benchmarking. A practical enterprise implementation combining dense and lexical retrieval in our controlled setting remains a meaningful direction.	Enterprise documents contain both semantic information and exact business terminology.
Research Gap 2 – Complementary semantic and lexical retrieval	DPR demonstrates strong dense retrieval, while BM25 remains a standard lexical retrieval method. The literature also shows that retrieval quality strongly affects downstream generation. The proposed study tests whether combining the two improves evidence retrieval for enterprise queries.	Combining complementary retrieval signals may recover information missed by either strategy alone.
Research Gap 3 – Practicality versus advanced RAG complexity	Self-RAG, CRAG and GraphRAG improve or extend RAG through additional mechanisms, but they also introduce more complex architectures or processing. There is room for a simpler, reproducible enterprise framework that focuses on retrieval quality.	A B.Tech-scale system should be implementable, measurable and explainable while still addressing a real retrieval problem.
Research Gap 4 – Controlled enterprise evaluation	ARES and EK-RAG demonstrate the importance of systematic RAG evaluation. A project-specific comparison using the same enterprise documents, questions and generation model can isolate the effect of the retrieval strategy.	This allows a fair comparison between the baseline and proposed model.

7. Research Objectives – Directly Derived from the Research Gaps

Objective	Directly addresses
Objective 1: To develop a practical Hybrid RAG framework for enterprise knowledge question answering using dense semantic retrieval and BM25 lexical retrieval.	Research Gap 1
Objective 2: To combine Sentence Transformer embeddings with FAISS semantic search and BM25 keyword retrieval and investigate whether the combined retrieval improves relevant-context retrieval.	Research Gap 2
Objective 3: To implement a comparatively simple and reproducible enterprise RAG architecture using open-source components and Llama 3 as the generation model.	Research Gap 3
Objective 4: To compare the proposed Hybrid RAG with a dense-retrieval RAG baseline using the same enterprise document collections, questions and generation model.	Research Gap 4
Objective 5: To evaluate retrieval and generation quality using Context Precision, Context Recall, Faithfulness, Answer Relevance, Answer Correctness and response time.	Research Gap 4

8. Proposed Model / Framework

The proposed model is a Hybrid RAG pipeline. Enterprise PDFs are loaded and cleaned, then divided into chunks. Sentence Transformers convert the chunks into embeddings that are indexed in FAISS. The same chunks are indexed for BM25 lexical retrieval. For a user query, both retrieval branches return candidate chunks. Their results are combined/ranked to form the final context, which is passed with the query to Llama 3 for grounded answer generation.

Execution flow: Enterprise Documents → PDF Loading → Cleaning → Chunking → Sentence Transformer Embeddings → FAISS Index + BM25 Index → User Query → Semantic Retrieval + Lexical Retrieval → Hybrid Result Fusion → Top-k Context → Llama 3 → Final Answer → Evaluation.

9. Datasets Proposed for the Current Project

The project will use three publicly available enterprise document collections. These PDFs are not conventional model-training datasets; they serve as the external knowledge base from which the RAG system retrieves evidence.

Collection	Planned documents	Official source
Microsoft	Annual reports and related investor/corporate reports	https://www.microsoft.com/en-us/investor/annual-reports
NVIDIA	Annual reports and related investor/corporate reports	https://investor.nvidia.com/financial-info/annual-reports-and-proxies/default.aspx
Infosys	Annual reports, ESG and corporate governance reports	https://www.infosys.com/investors/reports-filings/annual-report/annual-reports.html

10. Innovation

- Combines semantic and lexical retrieval in one enterprise RAG workflow.
- Uses the same enterprise document chunks for both retrieval branches, enabling controlled comparison.
- Separates retrieval evaluation from generation evaluation to identify where improvements occur.
- Uses multiple corporate document collections to test robustness across different enterprise content.

11. Creativity

- Provides a natural-language interface for asking questions about large collections of corporate documents.
- Can present the retrieved evidence alongside the generated answer for better transparency.
- Can be extended with source citations, document/page references, role-based access and continuous document updates.

12. Novelty

The novelty is not claimed as the invention of RAG, BM25, FAISS or semantic search. The proposed contribution is the practical integration and controlled evaluation of dense semantic retrieval and BM25 lexical retrieval for enterprise knowledge management, using multiple corporate document collections and a common LLM-based generation setup.

13. Feasibility

- Public enterprise reports are available through official company reporting/investor pages.
- The implementation can be developed in Python using LangChain, PyMuPDF, Sentence Transformers, FAISS, BM25 and Streamlit.
- Llama 3 can be used as the generation component without training an LLM from scratch.
- The project can begin with a manageable number of reports and a controlled question set, then scale if computing resources permit.
- RAG-specific evaluation metrics are available through established evaluation frameworks and can be complemented by manual validation.

14. Project Plan – Complete Execution Workflow

Phase 1 – Literature review: Review the selected papers; record model, outcome, limitations, datasets and evidence for each limitation.

Phase 2 – Research-gap synthesis: Group recurring limitations into specific Research Gaps and map each gap to one or more objectives.

Phase 3 – Dataset collection: Download and organize Microsoft, NVIDIA and Infosys enterprise reports from official sources.

Phase 4 – Document processing: Extract PDF text, clean extraction noise and split documents into meaningful chunks.

Phase 5 – Baseline RAG: Build dense retrieval using Sentence Transformers + FAISS and connect the retriever to Llama 3.

Phase 6 – BM25 retrieval: Build a BM25 index over the same chunks and test keyword-based retrieval.

Phase 7 – Hybrid retrieval: Combine/rank semantic and BM25 candidates and construct the top-k context.

Phase 8 – Answer generation: Pass the user query and retrieved context to Llama 3 and generate a grounded answer.

Phase 9 – Evaluation set: Create questions with reference answers and source evidence from the three enterprise collections.

Phase 10 – Evaluation: Measure Context Precision, Context Recall, Faithfulness, Answer Relevance, Answer Correctness and response time.

Phase 11 – Comparison: Compare dense RAG and Hybrid RAG under the same documents, questions, top-k setting and generation model.

Phase 12 – Demonstration and report: Develop a Streamlit interface, analyze results, document limitations/future scope and prepare the final report and presentation.

15. Expected Outcome

The expected outcome is an enterprise knowledge assistant that retrieves relevant evidence from corporate documents and generates grounded answers. The experiment will determine whether the proposed Hybrid RAG provides better retrieval and answer quality than the dense-retrieval baseline under controlled conditions.

16. References

1. Lewis, P. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. <https://arxiv.org/abs/2005.11401>
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